

Thesis Proposal Form

This form must be approved before a student can register for the dissertation. It is the student's responsibility to find and collaborate with a Mentor/supervisor to complete this form. Additionally, the student must ensure that all academic and administrative requirements are met. The form should be signed by the relevant parties, and submitted to the Dissertation Coordinator for final approval.

SECTION A: STUDENTS DETAILS

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Program: Computer Science



SECTION B: PROPOSAL

Proposed Dissertation Title:

AI Based Detection of Intracranial Aneurysms Using Deep Learning on Multi Modal Medical Imaging.

Research Question or Hypothesis:

The purpose of this research is to investigate how deep learning models can be used to detect intracranial aneurysms from multi modal medical imaging data while achieving high accuracy and reliability across different imaging modalities. Intracranial aneurysms are serious vascular abnormalities in the brain that can lead to life threatening conditions such as hemorrhagic stroke if not detected early. Accurate and early detection is therefore critical in clinical practice.

This research aims to answer the following question:

Which deep learning model provides the most accurate and reliable detection of intracranial aneurysms across CTA, MRA, and MRI imaging modalities?

The study will compare multiple deep learning models, including Convolutional Neural Networks and Residual Neural Networks, and evaluate their predictive performance using classification metrics such as AUC-ROC, accuracy, precision, and recall. Special emphasis will be placed on determining which model performs best in terms of both detection accuracy and generalization across different imaging modalities.

H0 (Null Hypothesis): There is no statistically significant difference in detection accuracy between baseline Convolutional Neural Network (CNN) models and advanced Residual Neural Network (ResNet) models when applied to intracranial aneurysm detection across CTA, MRA, and MRI imaging modalities.

H1 (Alternative Hypothesis): Advanced deep learning architectures, particularly ResNet models, will achieve statistically significantly higher detection accuracy than baseline CNN models in detecting intracranial aneurysms, due to their ability to learn deeper and more complex feature representations from multi-modal medical imaging data.

Review of Literature/Background:

Intracranial aneurysms are abnormal bulges in cerebral blood vessels that pose a serious health risk due to their potential rupture, which can result in subarachnoid hemorrhage, stroke, or death. It is estimated that a significant portion of the population may have unruptured aneurysms that remain undetected until severe complications occur. Early detection is therefore essential for reducing mortality rates and improving patient outcomes.

Traditionally, aneurysm detection is performed by radiologists using imaging techniques such as Computed Tomography Angiography, Magnetic Resonance Angiography and Magnetic Resonance Imaging. While these imaging modalities provide detailed visualization of brain vasculature, interpretation is time consuming and highly dependent on human expertise. Small or complex aneurysms may be difficult to identify, especially in high volume clinical environments.

Recent advancements in artificial intelligence, particularly deep learning, have significantly improved medical image analysis. Convolutional Neural Networks have demonstrated strong capabilities in automatically extracting spatial features from imaging data without manual feature engineering. More advanced architectures such as Residual Networks improve performance by enabling deeper networks through skip connections, which help overcome vanishing gradient problems.

Previous studies have shown that deep learning models can achieve radiologist level performance in several medical imaging tasks. However, challenges still exist in applying these models to intracranial aneurysm detection. These include variability in imaging modalities, differences in scanner quality, class imbalance in datasets, and limited generalization across institutions.

The RSNA Intracranial Aneurysm Detection dataset provides a valuable resource for addressing these challenges. It includes multi modal imaging data (CTA, MRA, MRI) with expert annotations indicating aneurysm presence and location. This dataset reflects real world clinical variability, making it suitable for evaluating the robustness of deep learning models.

This research contributes to existing literature by performing a comparative evaluation of multiple deep learning architectures on a multi modal dataset. Unlike previous studies that focus on single model performance, this study emphasizes comparative analysis and generalization across imaging modalities to better understand real world clinical applicability.

Research Methodology:

This study will adopt a quantitative experimental research methodology using the RSNA Intracranial Aneurysm Detection dataset. The dataset consists of anonymized medical imaging files in DICOM format, including CTA, MRA, and MRI scans, along with labeled annotations indicating aneurysm presence and location.

The research process will involve several stages.

First, the dataset will be preprocessed to convert DICOM images into a usable numerical format. Image preprocessing will include resizing, normalization, and standardization of pixel

intensity values. Since medical imaging datasets often suffer from class imbalance, techniques such as weighted loss functions and oversampling of minority classes will be applied to improve model learning.

Second, two deep learning models will be developed and evaluated. The first model will be a baseline Convolutional Neural Network, which will serve as a reference for performance comparison. The second model will be a Residual Neural Network using transfer learning with pre trained weights from large scale image datasets. This approach is expected to improve feature extraction and generalization performance.

Third, the models will be trained using supervised learning techniques. The dataset will be split into training, validation, and testing sets to ensure unbiased evaluation. Optimization will be performed using the Adam optimizer, and hyperparameter tuning will be applied to improve model convergence. Data augmentation techniques such as rotation, flipping, and scaling will also be used to improve robustness.

Fourth, the models will be evaluated on both binary classification (aneurysm present or absent) and multi-label classification tasks corresponding to aneurysm locations. Performance will be measured using evaluation metrics such as Area Under the Receiver Operating Characteristic Curve (AUC-ROC), accuracy, precision, recall, and F1-score, with AUC-ROC considered the primary evaluation metric.

Finally, comparative analysis will be conducted to determine which model performs best across different imaging modalities. Error analysis will also be performed to examine false positives and false negatives, particularly in cases involving small or difficult to detect aneurysms. The results will be interpreted in the context of clinical applicability and potential integration into radiological workflows. Litjens, G. et al. (2017) – A survey on deep learning in medical image analysis. Medical Image Analysis.

Selected Bibliography:

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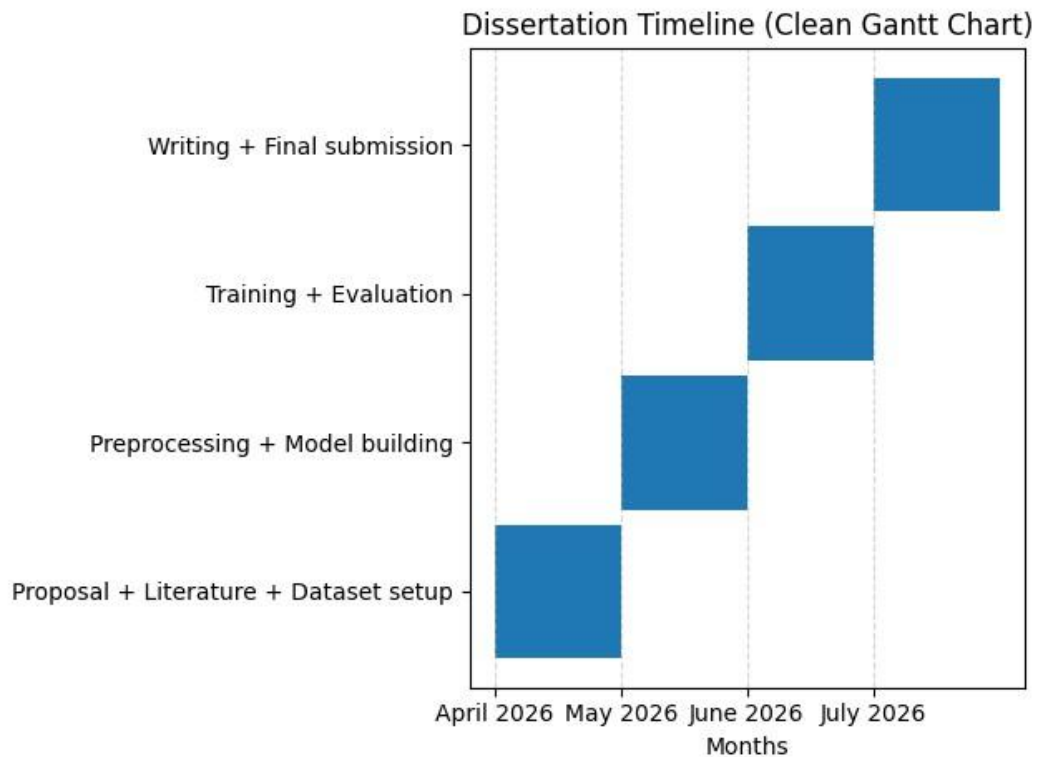
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Research and Writing Timeline:



Ethics:

This research uses publicly available anonymized medical imaging data from the RSNA dataset. The dataset does not contain any personally identifiable information. Therefore, no human participants are directly involved in this study. All data will be used strictly for academic purposes, and proper citations will be provided for all sources and datasets used.

The study follows academic integrity guidelines, ensuring transparency in methodology, honest reporting of results, and avoidance of plagiarism. The research also acknowledges that artificial intelligence models are intended to assist clinical decision making rather than replace medical professionals. Consideration will be given to potential biases and limitations in model performance across different imaging modalities and patient populations.

SECTION C: APPROVAL

Role	Name	Signature	Date
Mentor/Supervisor			
Dissertation Coordinator			
Academic Department			